

Lab 10 Answers

Yanqiao Chen*

Department of Computer Science and Engineering
Southern University of Science and Technology
12412115@mail.sustech.edu.cn

May 13, 2026

1 Question 1

We show that $\overline{EQ_{CFG}}$ is Turing-recognizable. We construct the following recognizer.

$D =$ “On input $\langle G_1, G_2 \rangle$, where G_1 and G_2 are CFGs:

- Enumerate all strings in Σ^* in lexicographic order, and for each string w , do the following:
 - Run the Turing machine for A_{CFG} on input $\langle G_1, w \rangle$ and the Turing machine for A_{CFG} on input $\langle G_2, w \rangle$ in parallel.
 - If one of the two Turing machines accepts and the other rejects, then accept.

If they are different, then there must exist a string w that is generated by one of the two CFGs but not the other. In this case, D will accept. If they are the same, then they may loop or reject, but they will never accept. Therefore, D is a recognizer for $\overline{EQ_{CFG}}$.

Thus, $\overline{EQ_{CFG}}$ is Turing-recognizable, thus EQ_{CFG} is co-Turing-recognizable.

2 Question 2

No, it does not. Let $f(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$. Thus let $B = \{1\}$. Then f is a

reduction from A to B . However, let $A = \{1^k 0^k \mid k \geq 0\}$, which is not regular, and $B = \{1\}$, which is regular. Therefore, A is not regular, but B is regular. Thus, the regularity of B does not imply the regularity of A .

3 Question 3

We know A is decidable $\iff A$ is Turing-recognizable and $\overline{A}$ is Turing-recognizable. Since A is Turing-recognizable, we only need to show that $\overline{A}$ is Turing-recognizable and $A \leq_m \overline{A}$. Thus, $f(x) \in \overline{A}$ if and only if $x \in A$ and

$f(x) \in A$ if and only if $x \in \bar{A}$. We construct the following Turing machine to recognize $\bar{A}$.

$D =$ “On input w , where w is a string:

- For any input, D run the Turing machine to map w to $f(w)$, and then run the Turing machine for A on input $f(w)$.
- If the Turing machine for A accepts, then reject; if it rejects, then accept.

Thus, $\bar{A}$ is Turing-recognizable. Therefore, A is decidable.

4 Question 4

Let M' be a Turing machine.

$M' =$ “On input $\langle x \rangle$, where x is a string:

- If $x = 01$, accept.
- Then run M on input w , if M accepts, then accept; if M rejects, then reject.

This construction perform a mapping reduction $f(\langle M, x \rangle) = \langle M' \rangle$. If the decider decides M' whether is accept w if and only if it accepts w^R , then if $x \neq 01$, then M' accepts x if and only if M accepts w , which means that it accept both w and w^R . Thus, if a decider can decide $M' \in T$, then it must decide whether M' accept w^R , which include 10, thus it must decide whether M accepts w , which is equivalent to deciding A_{TM} .

Thus, if M' is decidable, then the decider decides A_{TM} , which is a contradiction. Therefore, T is undecidable.

5 Question 5

We prove from right to left. If $A \leq_m A_{TM}$, we construct the following Turing machine to decide A .

$D =$ “On input w , where w is a string:

- Run the Turing machine to map w to $f(w)$, and then run the Turing machine M on input $\langle M, f(w) \rangle$. Then run the Turing machine for A_{TM} on input $\langle M, f(w) \rangle$ to recognize whether M accepts $f(w)$.
- If the Turing machine for A_{TM} accepts, then accept; if it rejects, then reject.

We prove from left to right. If A is decidable, we construct the following Turing machine to perform a mapping reduction from A to A_{TM} . As A is T-recognizable, there exists a Turing machine M that recognizes A . Let $f(x) = \langle M, x \rangle$, then we construct the following Turing machine to perform a mapping reduction from A to A_{TM} :

$D =$ “On input $\langle M, w \rangle$, where w is a string, M is a Turing machine:

- Simulate the Turing machine for A on input w . If it accepts, then accept
- If it rejects, then reject.

Thus $A \leq_m A_{TM}$, which completes the proof.

6 Question 6

Construct a mapping reduction from A_{TM} to T .

$$f(\langle M, w \rangle) = \langle M' \rangle$$

where M' is a Turing machine defined as follows:

$M' =$ "On input x , where x is a string:

- If $x \neq \epsilon$, accept.
- Then run M on input w , if M halts on w , accept; otherwise, reject.

Thus, if M' can be decided, then we can decide whether M accepts w , which is equivalent to deciding $HALT_{TM}$. Therefore, T is undecidable.

7 Question 7

- Undecidable. " $L(M)$ is infinite" is non-trivial, thus it is undecidable by Rice's theorem.
- Undecidable. " $1011 \in L(M)$ " is non-trivial, thus it is undecidable by Rice's theorem.
- Undecidable. " $L(M) = \Sigma^*$ " is non-trivial, thus it is undecidable by Rice's theorem.

8 Question 8

- True.
- False.
- True.
- False.
- False.
- True.

9 Question 9